COMPLETE AI LANGUAGES NOTES

Artificial Intelligence • Python • Machine Learning • Deep Learning • NLP • Computer Vision

A beginner-to-advanced study guide with concepts, Python examples, algorithms, workflows, evaluation methods, and practice projects.

Learning path: Programming fundamentals → mathematics and data handling → AI foundations → machine learning → deep learning → specialized AI → deployment and responsible AI.

Language focus: Python is the primary language for modern AI learning. Other useful languages and tools are included for context.

How to Use These Notes

These notes cover the major concepts and practical skills commonly meant by “AI languages.” AI is not one programming language: it is a field built using programming languages, libraries, data, algorithms, and computing infrastructure. Python is the main practical language in this guide.

• Read chapters in order if you are a beginner; experienced programmers can skim Python and mathematics.

• Run code examples in a notebook or Python file and change inputs to test your understanding.

• Treat code as educational examples: real projects need validation, error handling, security, testing, and monitoring.

• Build the projects near the end rather than only memorizing definitions.

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1. Artificial Intelligence: Overview

What is AI?

Artificial Intelligence (AI) is the study and engineering of systems that perform tasks associated with capabilities such as perception, prediction, language understanding, planning, and decision support. AI systems may use rules, search, statistical models, neural networks, or combinations of these.

Important terms

Term	Meaning
AI	Broad field of building systems that perform intelligent tasks.
Machine Learning (ML)	Methods that learn patterns from data instead of relying only on hand-written rules.
Deep Learning (DL)	Machine learning based on neural networks with multiple layers.
Data science	Collecting, cleaning, analyzing, and interpreting data; often overlaps with ML.
NLP	Natural Language Processing: working with human language.
Computer vision	Methods for interpreting images and video.
Generative AI	Models that create new text, images, audio, video, or other content.
Inference	Using a trained model to produce a prediction or output.
Training	Adjusting model parameters using examples and an optimization method.

Rule-based AI versus learning-based AI

A rule-based system follows explicit logic written by people. A learning-based system estimates patterns from examples. Rule-based methods can be suitable when rules are stable and explainable; learning-based methods can be useful when patterns are complex and enough representative data is available.

```
Rule-based example:
if temperature > 38:
    print('Possible fever')

Learning-based example:
model.fit(training_features, training_labels)
predictions = model.predict(new_features)
```

2. Programming Languages and the AI Ecosystem

Language/tool	Common role in AI
Python	Main language for data science, ML, deep learning, experimentation, and model APIs.
SQL	Querying, joining, aggregating, and managing structured data.
R	Statistics, data analysis, and research workflows.
C++	Performance-critical libraries, inference engines, and systems programming.
Java / Scala	Enterprise systems, data engineering, and JVM-based platforms.
JavaScript / TypeScript	AI features in web apps and browser-based experiences.
Julia	Numerical and scientific computing in some research settings.
CUDA	GPU programming ecosystem used to accelerate numerical workloads on supported NVIDIA hardware.
Bash / shell	Automation, environment setup, and running pipelines.

Recommended beginner stack: Python, a notebook environment, NumPy, pandas, Matplotlib, scikit-learn, and later PyTorch or another deep-learning framework. Add SQL when working with real datasets.

Typical workflow

• Define the problem and success metric.

• Collect data and check permissions and quality.

• Explore and split data into training, validation, and test sets.

• Build a simple baseline.

• Train and tune a model.

• Evaluate on data not used for fitting.

• Deploy, monitor, and improve responsibly.

3. Python Essentials for AI

Variables and basic types

```
age = 21                 # int
price = 19.95             # float
name = 'Asha'             # str
is_ready = True           # bool
scores = [82, 91, 76]     # list
profile = {'city': 'Warangal', 'active': True}  # dict
```

Conditions, loops, and functions

```
score = 87
if score >= 90:
    grade = 'A'
elif score >= 80:
    grade = 'B'
else:
    grade = 'C'

for value in [2, 4, 6]:
    print(value * 2)

def mean(values):
    if not values:
        raise ValueError('values must not be empty')
    return sum(values) / len(values)
```

Comprehensions, exceptions, and files

```
squares = [n * n for n in range(10)]

try:
    number = float('3.14')
except ValueError:
    print('Invalid number')

with open('notes.txt', 'w', encoding='utf-8') as file:
    file.write('AI practice')
```

Use clear names, small functions, docstrings for public functions, type hints where helpful, and tests for important logic. Avoid using eval on untrusted input.

Classes and objects

```
class Counter:
    def __init__(self):
        self.value = 0

    def increment(self):
        self.value += 1

counter = Counter()
counter.increment()
print(counter.value)  # 1
```

Useful Python habits

• Use virtual environments to isolate dependencies.

• Pin or record dependency versions for reproducibility.

• Separate data loading, preprocessing, training, and evaluation.

• Set random seeds where supported, while remembering some operations can remain nondeterministic.

• Do not store passwords, API keys, or private data in source code or public notebooks.

4. Data Structures, Functions, Modules, and Environments

AI programs still depend on ordinary programming foundations. Lists and arrays hold sequences; dictionaries map keys to values; sets track unique items; queues support work pipelines; trees and graphs represent hierarchies and relationships.

Structure	Typical use	Common operation notes
List/array	Ordered values, batches, features	Indexing is typically O(1) for arrays; insertion in the middle can be O(n).
Dictionary/hash map	Lookup by key, counters	Average lookup is often O(1), depending on implementation and hashing.
Set	Unique values and membership	Average membership testing is often O(1).
Queue	Task processing and breadth-first search	FIFO: first in, first out.
Heap/priority queue	Choosing highest- or lowest-priority item	Peek is O(1); insert/remove-top are usually O(log n).

Modules are reusable Python files; packages group modules. Environments isolate package versions. A notebook is useful for exploration, while production systems usually need tested modules, configuration, logging, and deployment scripts.

5. Mathematics for AI

Linear algebra

A scalar is a single number, a vector is an ordered set of numbers, a matrix is a 2-D array, and a tensor generalizes arrays to more dimensions. A dataset is often represented as a feature matrix X and target vector y.

```
import numpy as np
X = np.array([[1.0, 2.0], [3.0, 4.0]])
w = np.array([0.5, -0.2])
print(X @ w)  # matrix-vector multiplication
```

Dot products combine corresponding vector values and sum them. Matrix multiplication supports linear transformations. Vector norms measure magnitude. Eigenvectors and eigenvalues describe special directions and scaling of linear transformations.

Probability and statistics

Probability describes uncertainty. Mean summarizes central tendency; median is more robust to extreme values; variance and standard deviation describe spread. Conditional probability is P(A|B), the probability of A given B. Bayes' rule updates beliefs using evidence.

```
P(A_given_B) = P(B_given_A) * P(A) / P(B)  # when P(B) > 0
```

Distributions such as Bernoulli, binomial, normal, and categorical model different kinds of random outcomes. Correlation measures association, not necessarily causation.

Calculus and optimization

Derivatives describe local change. Gradients collect partial derivatives with respect to parameters. Gradient descent updates parameters in the direction that reduces a loss:

```
parameter = parameter - learning_rate * gradient
```

A learning rate that is too high can make training unstable; one that is too low can make it slow. In neural networks, backpropagation efficiently computes gradients using the chain rule.

Loss functions

• Mean squared error (MSE): common for regression.

• Binary cross-entropy: common for binary classification.

• Categorical cross-entropy: common for multi-class classification.

• Ranking and contrastive losses: useful in selected retrieval and representation-learning tasks.

6. Data Preparation and Visualization

Data quality checklist

• Check row counts, column types, missing values, duplicates, and impossible values.

• Understand what one row represents and how labels were created.

• Look for class imbalance, measurement errors, time effects, and sampling bias.

• Check whether the data can legally and ethically be used for the intended purpose.

Missing values and scaling

Missingness may be handled by dropping rows or columns when justified, or by imputing values. Fit imputers and scalers on training data only, then apply the learned transformation to validation/test data to avoid leakage.

```
from sklearn.model_selection import train_test_split
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# In a real workflow, fit preprocessing only on X_train.
```

Visualization

Histograms show distributions; scatter plots reveal relationships; box plots help spot unusual values; line charts show time trends; confusion matrices summarize classification errors. Labels, units, sample sizes, and uncertainty should be clear.

7. AI Problem Types and End-to-End Workflow

Problem	Input	Output example
Regression	Features	Numeric value such as demand
Classification	Features	Class such as spam/not spam
Clustering	Unlabelled examples	Groups of similar customers
Ranking/retrieval	Query and candidate items	Relevant results ordered by score
Generation	Prompt/context	New text, image, audio, or code
Control/decision	State and actions	Action selected to maximize long-term reward

Start with a clear problem statement, baseline, metric, and data split. Determine what errors cost most. The best model is not always the largest model: latency, cost, privacy, interpretability, maintainability, and robustness matter too.

8. Machine Learning Foundations

Supervised, unsupervised, and reinforcement learning

• Supervised learning uses labelled examples (features X and target y).

• Unsupervised learning looks for structure without target labels, such as clusters or lower-dimensional representations.

• Reinforcement learning trains an agent to choose actions through interaction and reward signals.

Generalization and overfitting

A model generalizes when it performs well on new examples. Overfitting occurs when it learns training-specific details that do not transfer; underfitting occurs when it fails to capture important patterns. More model complexity is not automatically better.

Baseline and data leakage

A baseline provides a simple performance reference, such as predicting the majority class or mean target. Data leakage occurs when information unavailable at prediction time influences training or evaluation. Leakage can make a model appear much better than it really is.

Scikit-learn example

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

X, y = load_iris(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
model = make_pipeline(StandardScaler(), LogisticRegression(max_iter=1000))
model.fit(X_train, y_train)
pred = model.predict(X_test)
print(accuracy_score(y_test, pred))
```

9. Supervised Learning Algorithms

Algorithm	Core idea	Strengths / cautions
Linear regression	Fit a linear relationship to numeric targets	Simple, interpretable baseline; sensitive to assumptions and outliers.
Logistic regression	Model class probabilities through a logistic link	Strong baseline; linear decision boundary unless features are transformed.
Decision tree	Split data using feature-based rules	Interpretable; can overfit without constraints.
Random forest	Average many randomized trees	Often robust; less directly interpretable.
Gradient boosting	Build an ensemble sequentially to improve errors	Powerful on tabular data; tuning matters.
k-nearest neighbors	Predict from nearby training examples	Simple; scaling and distance choice matter.
Support vector machine	Find a margin-based decision boundary	Effective in some high-dimensional settings; scaling and kernel choice matter.
Naive Bayes	Apply Bayes' rule with conditional-independence assumptions	Fast baseline for text; assumptions may be unrealistic.

Regression versus classification

Regression predicts a quantity. Classification predicts a category or a probability for each category. For imbalanced classes, accuracy alone can be misleading; examine precision, recall, F1, and the confusion matrix.

Feature engineering

Feature engineering turns raw data into useful predictors: date parts, counts, domain-specific ratios, text representations, or interaction terms. Features must be available at prediction time and computed without leaking future information.

10. Unsupervised Learning and Dimensionality Reduction

Clustering

K-means assigns examples to k clusters by iteratively updating cluster centres. It is efficient but sensitive to scale, initialization, and the choice of k; it works best when cluster geometry is reasonably compatible with its distance objective.

```
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
X_scaled = StandardScaler().fit_transform(X)
clusters = KMeans(n_clusters=3, random_state=42, n_init='auto').fit_predict(X_scaled)
```

Hierarchical clustering builds nested groups. DBSCAN identifies density-connected groups and can mark noise, but depends on distance and density parameters.

Dimensionality reduction

Principal Component Analysis (PCA) finds orthogonal directions that capture high variance in the data. It is often used for compression, visualization, or denoising. PCA is not a feature-selection method and does not guarantee that low-variance features are unimportant for a target.

Anomaly detection

Anomaly detection flags examples that differ from expected patterns. Methods include statistical thresholds, isolation forests, one-class SVMs, and autoencoders. Evaluate alerts with domain context because rare valid cases may look unusual.

11. Model Evaluation and Validation

Classification metrics

Metric	Meaning
Accuracy	Correct predictions / all predictions; may mislead under class imbalance.
Precision	Of predicted positives, fraction that are truly positive.
Recall / sensitivity	Of actual positives, fraction detected.
F1 score	Harmonic mean of precision and recall.
ROC-AUC	Ranking ability across thresholds; can be less informative for very rare positives.
PR-AUC	Precision-recall trade-off; often helpful with rare positive classes.

Regression metrics

MAE averages absolute errors. MSE squares errors and penalizes large errors more strongly; RMSE is its square root and returns to the target's units. R^2 compares performance with a mean-prediction reference under its standard definition.

Splits and cross-validation

Training data fit model parameters; validation data supports tuning and model selection; test data estimates final performance. Cross-validation repeats training/validation across folds. For time-dependent data, use time-aware splits instead of random shuffling. Grouped data may require group-aware splitting.

Thresholds and calibration

A classifier's probability score is not automatically well calibrated. Choose thresholds according to real-world costs and constraints, and evaluate calibration when probability estimates drive decisions.

Error analysis

• Review false positives and false negatives separately.

• Measure performance across relevant subgroups where appropriate and lawful.

• Inspect data slices, edge cases, drift, and uncertainty.

• Compare to a simple baseline and report variability, not only a single score.

12. Neural Networks and Deep Learning

Neurons, layers, and activations

A neural network applies learned transformations to inputs. A simple layer computes a weighted sum plus bias, then an activation function. ReLU is common in hidden layers; sigmoid and softmax are often used for output probabilities in suitable settings.

Training loop concepts

• Forward pass: compute predictions.

• Loss: quantify disagreement between predictions and targets.

• Backpropagation: calculate gradients.

• Optimizer: update parameters.

• Batch: subset of training examples.

• Epoch: one pass over the training set.

• Regularization: methods such as weight decay, dropout, or data augmentation to improve generalization.

PyTorch-style model sketch

```
import torch
from torch import nn

model = nn.Sequential(
    nn.Linear(10, 32),
    nn.ReLU(),
    nn.Linear(32, 2)
)
# For classification, the output here is logits.
# Use an appropriate loss such as CrossEntropyLoss for class-index labels.
```

For multi-class classification with PyTorch CrossEntropyLoss, pass raw logits and integer class labels; do not apply softmax before the loss. During inference, switch to evaluation mode and disable gradient tracking when appropriate.

Common challenges

• Vanishing/exploding gradients; use suitable initialization and architectures.

• Overfitting; use validation, regularization, and adequate data.

• GPU memory limits; use smaller batches, mixed precision where safe, or smaller models.

• Reproducibility; record datasets, configuration, random seeds, code versions, and checkpoints.

13. Natural Language Processing and Large Language Models

Text processing

NLP systems may normalize text, split it into tokens, represent tokens as IDs or vectors, and learn contextual relationships. Traditional methods include bag-of-words and TF-IDF; modern systems commonly use learned embeddings and transformer models.

Transformers and attention

Attention lets a model combine information from different token positions. Transformer architectures use attention and feed-forward layers to build contextual representations. Positional information helps represent order.

Large language models (LLMs)

An autoregressive language model learns to predict the next token given previous context. It can be adapted through prompting, fine-tuning, or retrieval-augmented generation (RAG). The model generates plausible outputs, but plausibility is not the same as truth.

Prompting principles

• State the task, audience, constraints, and desired output format.

• Provide relevant context and representative examples when useful.

• Ask the model to distinguish facts, assumptions, and uncertainty.

• Validate important claims independently and test with edge cases.

• Do not include secrets or personal data unless the service and use case are approved for it.

Retrieval-augmented generation

RAG retrieves relevant documents and provides them to a language model as context. A typical pipeline is: ingest documents → chunk → embed → index → retrieve → optionally rerank → construct grounded prompt → generate → evaluate. RAG can improve grounding, but retrieval can fail and model outputs still require evaluation.

LLM evaluation

Assess task success, factuality, citation quality where applicable, safety, robustness, latency, cost, and user outcomes. Use representative test sets and human review for subjective or high-impact tasks. Do not rely on a single aggregate score.

14. Computer Vision

Computer vision processes images and video. Image classification assigns labels; object detection predicts locations and classes; segmentation labels pixels; OCR extracts text; tracking follows objects across frames.

Image pipelines

• Decode and validate image files; consider orientation and color channels.

• Resize and normalize consistently.

• Use augmentation only when transformations preserve the intended label.

• Split by person, video, location, or source where needed to prevent near-duplicate leakage.

• Measure performance on realistic deployment conditions, not only clean benchmark images.

Convolutional neural networks (CNNs) use local filters and shared weights. Vision transformers process image patches with attention. Transfer learning starts from a model trained on a large dataset and adapts it to a target task.

Computer vision metrics

Classification can use accuracy, precision, recall, and F1. Detection commonly uses intersection over union (IoU), average precision, and mean average precision. Segmentation can use IoU or Dice score. The correct metric depends on the task and error costs.

15. Reinforcement Learning

Reinforcement learning (RL) studies an agent interacting with an environment. A state describes the situation, an action is a choice, a reward gives feedback, and a policy maps states to actions. The goal is generally to maximize expected cumulative reward.

Key concepts

• Exploration: trying actions to learn; exploitation: using current knowledge.

• Value function: expected future return from a state or state-action pair.

• Discount factor: controls how future rewards are weighted.

• Q-learning: updates action-value estimates using observed rewards and future value.

• Policy gradient methods optimize a parameterized policy directly.

RL can be sample-inefficient and unstable. Start with small, safe simulated environments. In real-world systems, define constraints, safety checks, and offline evaluation before allowing learned policies to affect people or physical systems.

16. Generative AI and Prompt Engineering

Generative models learn data patterns and sample new outputs. Examples include language models for text, diffusion models for images, and generative audio models. Outputs can be wrong, biased, copyrighted, unsafe, or inconsistent with a user's intent.

Useful prompt structure

```
Task: Summarize the supplied report.
Audience: A non-technical manager.
Constraints: Use only information in the report; flag missing evidence.
Format: 5 bullets followed by 3 open questions.
Source material: [insert approved text]
```

Model adaptation choices

Method	Best suited for	Limitations
Prompting	Fast task steering without model updates	Sensitive to wording and context limits.
RAG	Answers grounded in a changing document collection	Needs good retrieval, access control, and source evaluation.
Fine-tuning	Teaching a model a stable style or task pattern	Requires curated examples, evaluation, and training resources.
Tool use	Calling search, databases, or calculators	Tools need permission boundaries and input/output validation.

Choose the simplest approach that meets the quality requirement. Fine-tuning does not automatically make a model factually current; RAG does not automatically prevent hallucinations; prompting does not guarantee compliance.

17. Data Pipelines, APIs, and Deployment

From notebook to service

• Package reusable code and model artifacts.

• Validate input schema, types, ranges, and size.

• Expose a clear interface, often an API or batch job.

• Log operational metadata without unnecessarily logging sensitive inputs.

• Use access controls, rate limits, monitoring, versioning, and rollback plans.

• Test latency, throughput, memory use, failure modes, and dependency vulnerabilities.

Simple API concept

```
from fastapi import FastAPI
from pydantic import BaseModel

app = FastAPI()

class PredictionInput(BaseModel):
    features: list[float]

@app.post('/predict')
def predict(payload: PredictionInput):
    # Validate feature length/ranges and call a loaded model here.
    return {'status': 'example endpoint; connect a trained model'}
```

The example illustrates API shape only; it is not a complete production prediction service. A real service should load a versioned model, validate feature dimensions, handle exceptions, and implement authentication where needed.

Monitoring

Monitor data drift, prediction distributions, quality metrics when labels arrive, latency, error rates, cost, and resource use. Retraining should be based on a reviewed process, not automatically triggered by every shift.

MLOps

MLOps combines software engineering and machine-learning practices: experiment tracking, data/model versioning, automated tests, reproducible pipelines, deployment, monitoring, and governance.

18. Responsible, Secure, and Reliable AI

• Privacy: minimize data collection, restrict access, protect sensitive information, and respect retention requirements.

• Fairness: evaluate meaningful differences in error rates and impacts; mitigation requires context and careful trade-offs.

• Transparency: explain system purpose, limitations, and the role of automation.

• Security: protect credentials, isolate tools, validate inputs, and treat retrieved documents as untrusted content.

• Human oversight: preserve appropriate review and appeal for consequential decisions.

• Robustness: test distribution shifts, adversarial inputs, malformed files, and failure cases.

• Copyright and consent: confirm rights and permissions for training data, prompts, and generated outputs.

• Safety: define prohibited uses, escalation paths, incident reporting, and rollback procedures.

For high-impact uses—such as health, finance, hiring, education, or public services—use qualified domain experts, applicable law and policy, documented risk assessment, and meaningful human review. AI output should not be treated as an unquestionable authority.

19. Other Languages and When to Use Them

SQL

```
SELECT category, COUNT(*) AS item_count, AVG(price) AS avg_price
FROM products
GROUP BY category
ORDER BY item_count DESC;
```

SQL is essential for querying structured datasets and is often used before Python receives data for analysis.

R

R is popular for statistical analysis, visualization, and reproducible research. It has strong statistical packages and a mature visualization ecosystem.

JavaScript / TypeScript

These languages integrate AI features into web applications, run inference in supported browser or server environments, and connect user interfaces to model APIs.

C++ and CUDA

C++ is useful when performance and low-level control are important. CUDA is used to write GPU kernels for compatible NVIDIA GPUs; most learners can start with Python frameworks and study lower-level optimization later.

Language choice summary

Start with Python for learning and prototyping, SQL for data access, and the language already used by your application team for integration. Add lower-level languages only when performance, platform constraints, or system requirements justify them.

20. Common AI Interview Questions

Q: What is the difference between AI, ML, and deep learning?
A: AI is the broad field; ML learns from data; deep learning is ML using multi-layer neural networks.

Q: What is overfitting?
A: When a model fits training-specific patterns that do not generalize well to unseen examples.

Q: Why split data?
A: To estimate generalization and separate model fitting, tuning, and final evaluation.

Q: What is data leakage?
A: Information from outside the permitted training process—often from the future or target construction—unfairly influences training or evaluation.

Q: Precision versus recall?
A: Precision measures correctness among predicted positives; recall measures the share of actual positives detected.

Q: What is regularization?
A: A family of techniques that constrain learning to reduce overfitting.

Q: What is gradient descent?
A: An optimization method that updates parameters using gradients of a loss function.

Q: Why scale features?
A: Some models and distance-based methods are sensitive to feature magnitude; scaling can improve optimization and fair distance comparisons.

Q: What is an embedding?
A: A learned dense vector representation of an item such as a word, sentence, image, or user.

Q: What is attention?
A: A mechanism that computes weighted combinations of representations based on relevance to a query.

Q: What is RAG?
A: Retrieval-augmented generation retrieves external context and supplies it to a generative model.

Q: Why is accuracy sometimes misleading?
A: A majority-class predictor can score highly when positive cases are rare.

Q: What is a baseline?
A: A simple reference system used to judge whether a more complex model adds value.

Q: What is model drift?
A: A change in data or relationships that can degrade model performance over time.

Q: What is inference latency?
A: The time required to produce an output for an input or request.

21. Exercises and Project Roadmap

Beginner exercises

• Write Python functions for mean, median, and standard deviation.

• Load a CSV, inspect missing values, and create two charts.

• Build a rule-based classifier and compare it with a learned model.

• Use train/test splitting and report precision, recall, and F1.

Intermediate projects

• House-price or demand regression with a documented baseline and error analysis.

• Spam or sentiment classification with a text pipeline.

• Customer segmentation with scaled features and a discussion of cluster limitations.

• Image classification using transfer learning and leakage-aware splits.

Advanced projects

• Document question-answering RAG system with citations and a retrieval evaluation set.

• Deploy a model API with input validation, versioning, monitoring, and tests.

• Compare small and large models on quality, cost, latency, privacy, and failure modes.

• Build an end-to-end pipeline with data validation, reproducible training, and model registry.

Suggested 12-week roadmap

Weeks	Focus	Deliverable
1–2	Python, functions, data structures, Git, environments	Small tested Python project
3–4	NumPy, pandas, charts, probability, linear algebra basics	Exploratory data analysis report
5–6	Supervised ML and evaluation	Reproducible tabular ML project
7–8	Unsupervised learning, feature engineering, validation	Clustering or anomaly project
9–10	Neural networks, NLP or vision fundamentals	Small deep-learning experiment
11	LLMs, prompting, RAG, responsible AI	Evaluated prototype
12	API deployment, monitoring, documentation	Demonstration-ready project

Project report template

• Problem and intended users

• Data source, permission, and limitations

• Baseline and evaluation design

• Model and preprocessing choices

• Metrics and error analysis

• Fairness, privacy, and safety considerations

• Deployment, monitoring, and rollback plan

• What did not work and what to try next

22. Glossary and Revision Checklist

Term	Quick definition
Feature	Input variable used by a model.
Label/target	Value the model is trained to predict.
Parameter	Value learned by a model during training.
Hyperparameter	Configuration chosen outside ordinary parameter fitting.
Epoch	One full pass through training examples.
Batch	Subset processed in one training step.
Embedding	Dense vector representation.
Token	A unit of text processed by a tokenizer.
Inference	Producing outputs using a model.
Drift	Change in data or relationships over time.
Calibration	Agreement between predicted probabilities and observed frequencies.
Reproducibility	Ability to repeat a process with recorded inputs and configuration.

Final revision checklist

• I can write and test basic Python functions and use modules and environments.

• I understand arrays, data frames, feature matrices, labels, and data cleaning.

• I can explain supervised, unsupervised, deep, and reinforcement learning.

• I can choose metrics that match the task and class balance.

• I understand leakage, overfitting, validation, and generalization.

• I can describe neural networks, transformers, embeddings, and RAG at a high level.

• I can design a reproducible workflow from data to deployment.

• I can identify privacy, fairness, security, and reliability risks.

• I can explain my model's limitations and support claims with measured evidence.

Closing note: AI is a broad and fast-moving field. Use these notes as a foundation, then consult current official documentation for the specific libraries, models, and platforms you choose. Build small experiments, measure results, and document limitations.